Problem Set 7

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Question 6: Summary Statistics and Missing Data

The proportion of missing values in the `logwage` variable is found to be around 25.1%. This is a significant level of missing data in the dataset. Considering that the information on wages might be sensitive and could be a function of some observable factors such as education, age, and working conditions, it is not probable that the missing data is Missing Completely At Random (MCAR). Rather, it is more probable that the data is Missing At Random (MAR), where the probability of missing data is a function of some observed variables.

	Unique	Missing Pct.	Mean	SD	Min	Median	Max	Histogram
logwage	670	25	1.6	0.4	0.0	1.7	2.3	
hgc	16	0	13.1	2.5	0.0	12.0	18.0	
tenure	259	0	6.0	5.5	0.0	3.8	25.9	
age	13	0	39.2	3.1	34.0	39.0	46.0	
		N	college	college grad	530	23.8		
	not college grad	1699	76.2					
married	married	1431	64.2					
	single	798	35.8					

Question 7: Imputation and Regression Analysis

The coefficient on `hgc` represents the return to schooling. Across the four imputation methods, the estimated values of $\hat{\beta}_1$ vary. The estimate obtained using the complete case and regression imputation methods is found to be similar, at 0.062, whereas the multiple imputation method provides a slightly lower estimate at 0.060. The mean imputation method provides the least estimate at 0.050.

All methods, apart from mean imputation, underestimate the true value of $\beta_1 = 0.093$. Mean imputation performs the worst, providing the most biased estimate, as it has a greater tendency to decrease the variability in the dependent variable. The complete case and regression imputation methods provide identical results, as regression imputation has a greater tendency to maintain the structure of the data, though it does not take into account the imputed values.

On the other hand, multiple imputation offers a more statistically robust method, where uncertainty is taken into consideration through multiple imputed datasets. Although its result is slightly lower than the complete case result, multiple imputation is generally preferred over other methods due to its theoretical advantages in handling missing data. All these results show that naive imputation techniques can significantly change the relationship in the data, while multiple imputation offers a more robust method for analysis.

	Complete Case	Mean Imputation	Regression Imputation	Multiple Imputat
(Intercept)	0.534	0.708	0.534	0.593
	(0.146)	(0.116)	(0.112)	(0.138)
hgc	0.062	0.050	0.062	0.060
	(0.005)	(0.004)	(0.004)	(0.005)
collegenot college grad	0.145	0.168	0.145	0.121
	(0.034)	(0.026)	(0.025)	(0.035)
tenure	0.050	0.038	0.050	0.044
	(0.005)	(0.004)	(0.004)	(0.005)
I(tenure^2)	-0.002	-0.001	-0.002	-0.001
	(0.000)	(0.000)	(0.000)	(0.000)
age	0.000	0.000	0.000	0.000
	(0.003)	(0.002)	(0.002)	(0.003)
marriedsingle	-0.022	-0.027	-0.022	-0.015
	(0.018)	(0.014)	(0.013)	(0.017)
Num.Obs.	1669	2229	2229	2229
Num.Imp.				5
R2	0.208	0.147	0.277	0.228
R2 Adj.	0.206	0.145	0.275	0.226
AIC	1179.9	1091.2	925.5	
BIC	1223.2	1136.8	971.1	
Log.Lik.	-581.936	-537.580	-454.737	
F	72.917	63.973	141.686	
RMSE	0.34	0.31	0.30	

Question 8: Project Progress

For this project, I plan to extend my previous project on linguistic diversity, which I conducted in my Data Visualization class, where I analyzed the relationship between linguistic diversity, digital access, and development outcomes in different countries. In this project, I used interactive visualizations to identify relationships between different variables, such as internet access, GDP per capita, linguistic diversity, and E-Government Development Index (EGDI). From this analysis, I found a strong positive relationship between digital access and development, as well as a weaker negative relationship between linguistic diversity and internet penetration in different countries.

This project provided me with some significant insights, but I found that this analysis was more descriptive in nature, without any formal statistical tests.

For this class, I plan to extend this project by incorporating econometric modeling techniques to formally analyze these relationships. In this project, I plan to use regression analysis, where digital access/development outcomes, such as internet usage and EGDI, are used as dependent variables, and linguistic diversity is a key independent variable, along with other independent variables such as GDP per capita. This will allow me to formally analyze these relationships and determine whether these relationships hold in a more rigorous analysis, as well as determine the extent to which linguistic diversity affects these outcomes. This extension will take my project a step further in aligning it with this class, where I will be using econometric techniques to analyze these relationships.